A3-PS033-TC440

PS-033

DNS-Based Discovery of Diagnostic Services

NIRIKSHA

“Discover. Examine. Verify.”

Team DARSHAN 2.0

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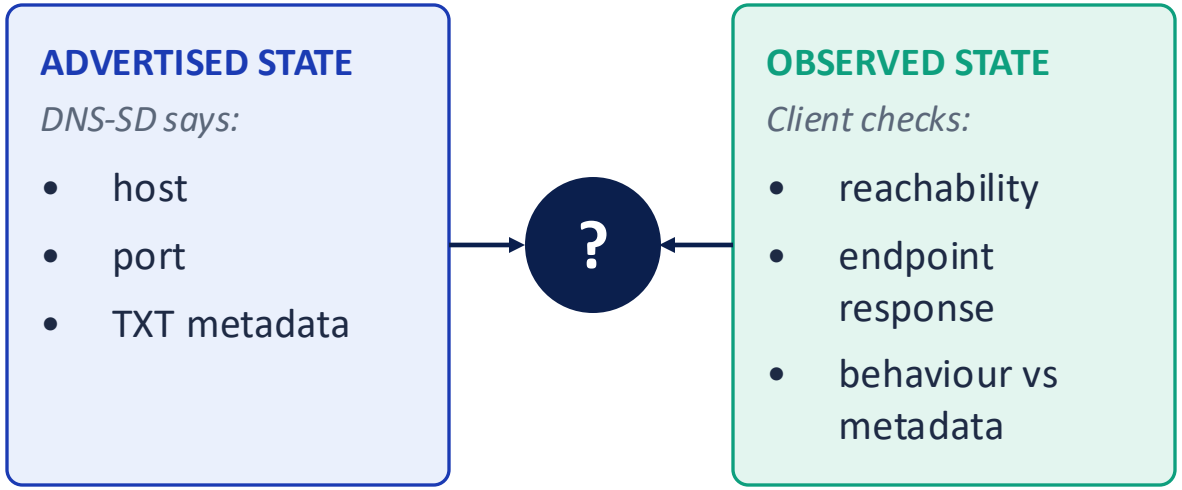
- **Blue wing —** advertised state (DNS-SD)
- **Green wing —** client-observed state
- **Iris —** independent verification
- **Node ring —** discovered services

Advertised state → Examination → Verified observation

Problem Description

Discovery tells us what is advertised. It does not by itself prove current usability.

1. Clients should not depend on manually configured service IPs and ports.
2. DNS-SD enables services to advertise and clients to dynamically discover available service instances.
3. A discovered endpoint may still be unavailable or misconfigured.
4. Advertised SRV / TXT data may not match current application behaviour.

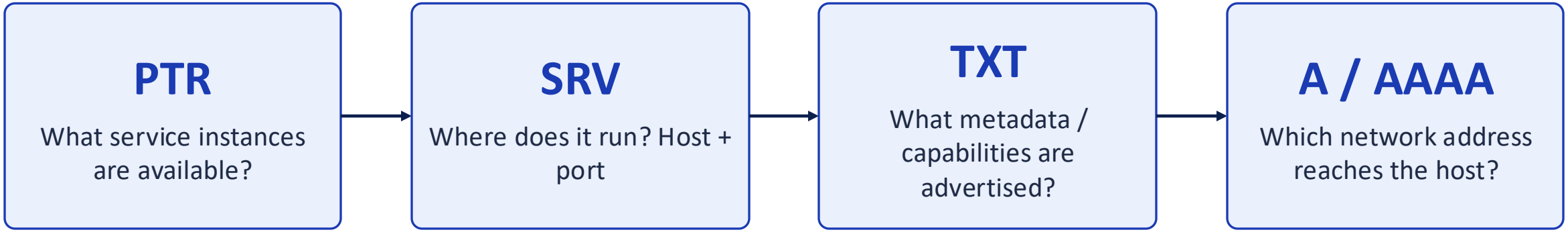


CENTRAL RESEARCH QUESTION

What was advertised, what did the client observe, and do the two agree?

Background

DNS-SD records describe a service. NIRIKSHA treats them as claims to verify.



ILLUSTRATIVE TXT PROFILE (application-specific)

```
diag=dns
txtvers=1
health=/health
run=/diagnose
```

DNS-SD solves service discovery.

NIRIKSHA asks the next operational question:

Is the discovered diagnostic service actually usable as advertised?

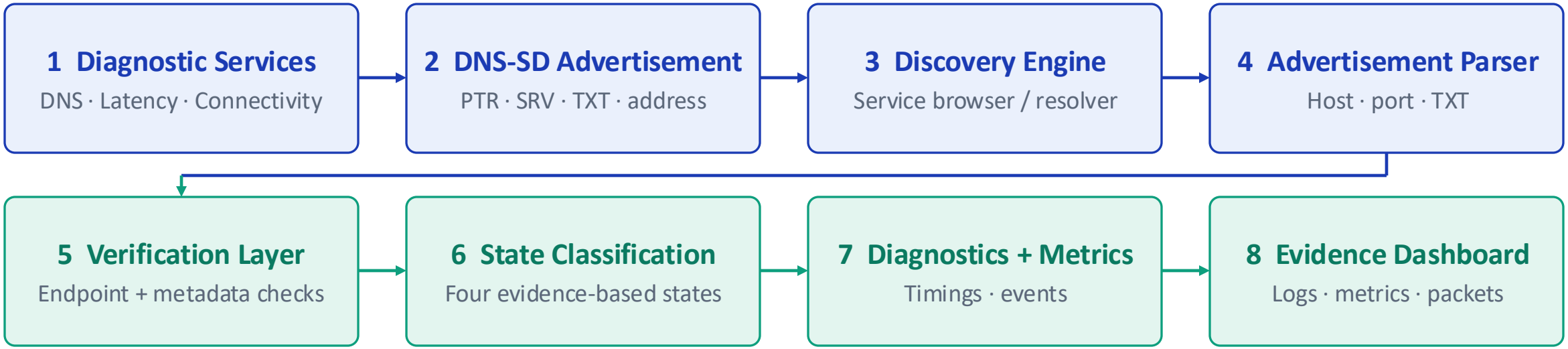
PTR / SRV / TXT per RFC 6763 [2]; link-local mDNS operation per RFC 6762[1]. The TXT keys are a NIRIKSHA prototype profile, not an IETF standard. IPv6 checks are experimental and environment-dependent; SRP (RFC 9665 [5]) is future scope.

Literature & RFC Review

Source / Standard	What it contributes	Observation relevant to NIRIKSHA
RFC 6762 · Multicast DNS [1]	Link-local name resolution and record discovery without a unicast DNS server.	Provides the local-link mDNS discovery mechanism used by the prototype and visible in packet captures.
RFC 6763 · DNS-SD [2]	Structures PTR, SRV and TXT records so clients can browse named service instances.	Defines the advertised state that NIRIKSHA treats as a claim to verify.
RFC 7558 · Scalable DNS-SD [3]	Problem statement and requirements for DNS-SD beyond a single link.	Discovery is an operational concern; NIRIKSHA examines client-side usability.
RFC 8766 · Discovery Proxy [4]	Makes link-local mDNS service records visible in the unicast DNS namespace.	DNS-SD reaches beyond one link; cross-subnet use is future scope here.
RFC 9665 · SRP [5]	Unicast service registration using DNS Update with SIG(0) keys.	Relevant future extension; not part of the committed prototype.

Positioning: NIRIKSHA does not claim to invent DNS-SD or ordinary health monitoring; it studies the advertised-vs-observed gap under controlled faults.

NIRIKSHA — Discover. Examine. Verify.

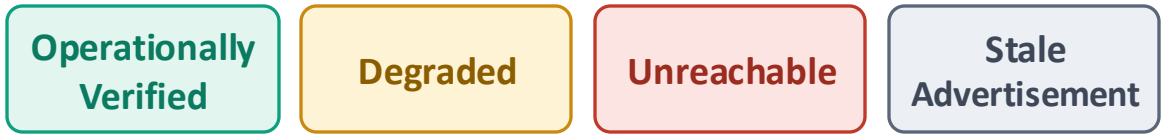


CONCEPTUAL TRANSFORMATION



Planned services: DNS / Latency / Connectivity
Output: evidence + events + measurements

SERVICE STATE CLASSIFICATION



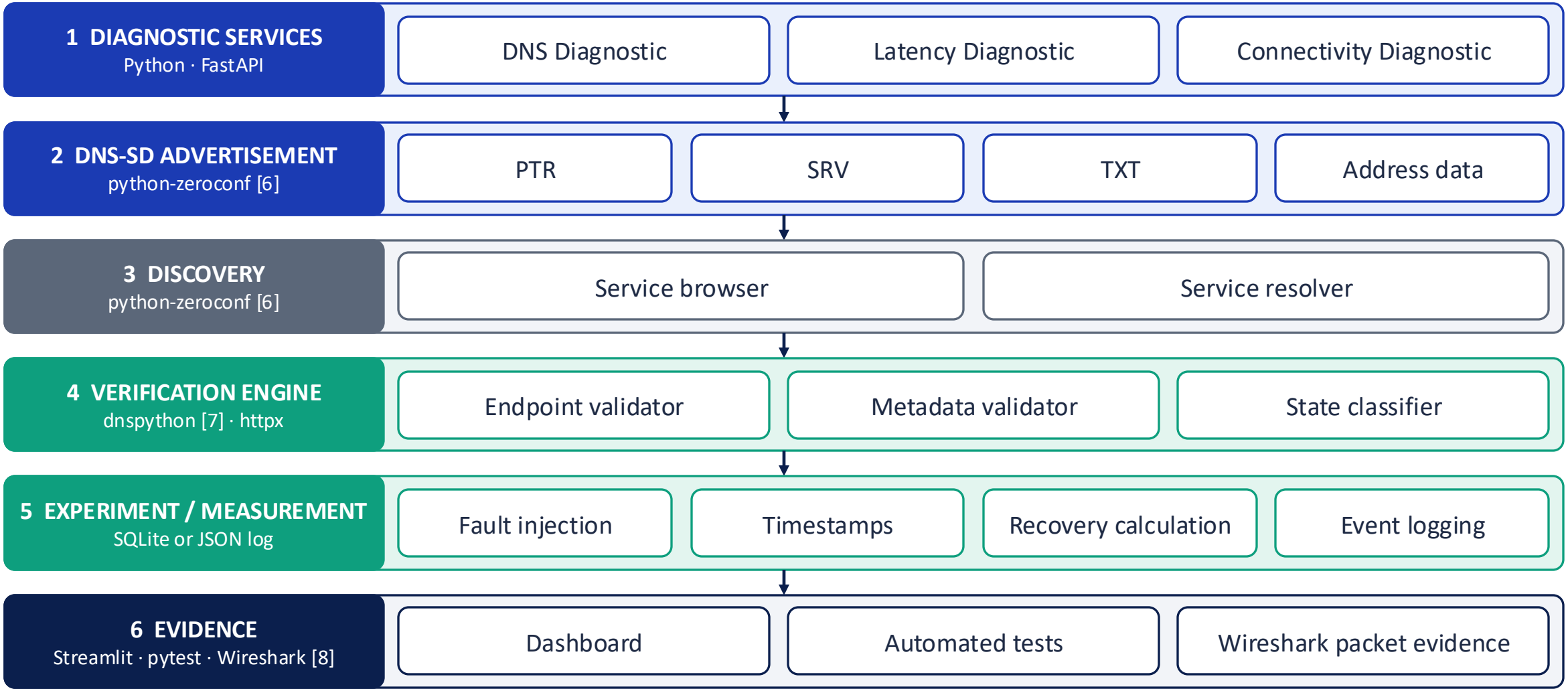
Optimization Proposed

Baseline: discovery-oriented workflow		Proposed: NIRIKSHA workflow
Hard-coded service endpoint	→	**Dynamic DNS-SD discovery**
Discovery-only view: service is listed	→	**Discovered + independently verified**
Advertised port shown directly	→	**Endpoint reachability verified**
TXT metadata displayed	→	**Selected metadata tested against behaviour**
Failures observed manually	→	**Controlled fault detection + event logging**
Endpoint change needs manual interpretation	→	**Replacement discovery + recovery timing**
Visual monitoring only	→	**Metrics + packet evidence + automated tests**

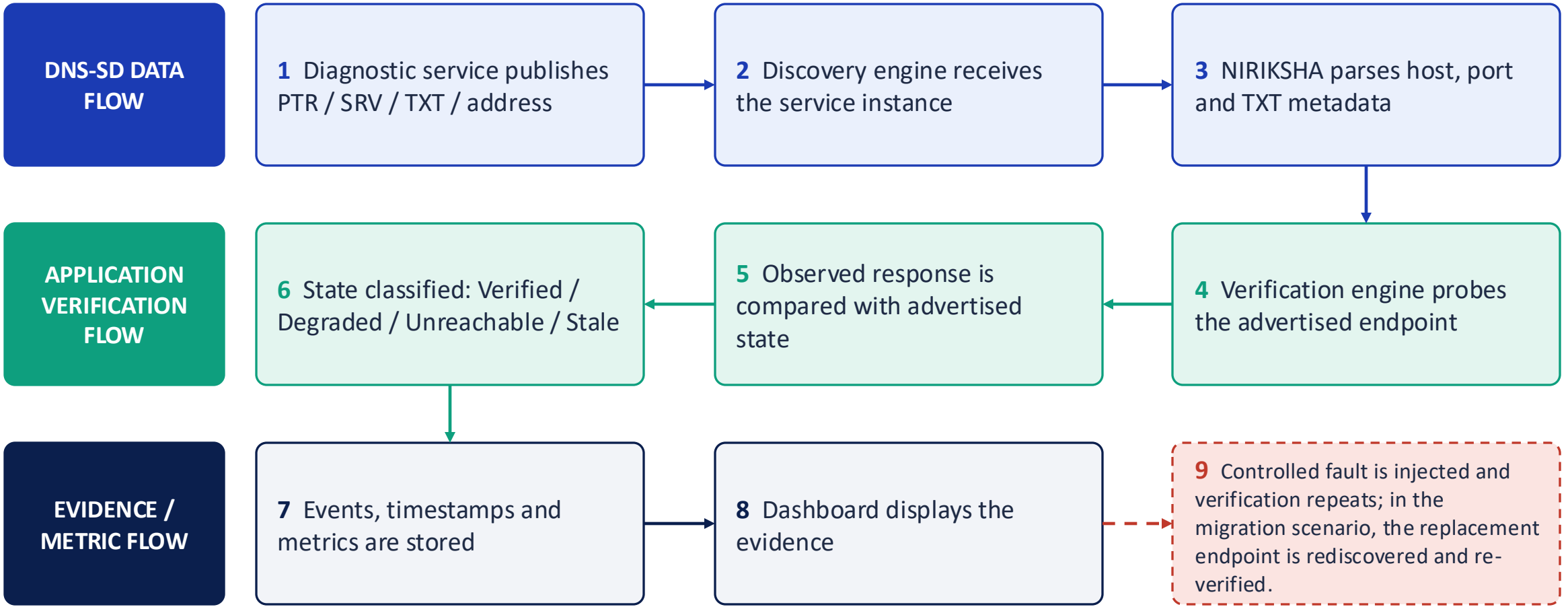
Framed as a baseline discovery-oriented workflow versus the proposed NIRIKSHA workflow — not a statement about all existing systems.

Solution Architecture & Design

■ Advertised side ■ Verification and evidence



Data Flow Diagram



blue = DNS-SD data flow **teal** = application verification flow **navy** = evidence / metric flow **red dashed** = controlled-fault loop

Deployment Timeline



DAY 1	DAY 2	DAY 3	DAY 4	DAY 5	DAY 6	DAY 7
BUILD Environment, one service, DNS-SD discovery	**BUILD** PTR / SRV / TXT / address handling	**BUILD** Three real diagnostic services	**BUILD** Verification engine and state classification	**BUILD** Fault injection; failure and replacement detection	**BUILD** Metrics, dashboard, tests, Wireshark evidence	**BUILD** End-to-end stabilization and documentation
GATE Discovery works with no manual IP or port	**GATE** PTR, SRV, TXT and address parsed correctly	**GATE** All three services return real results	**GATE** Wrong port and wrong TXT detected	**GATE** All four fault/migration scenarios detected and logged.	**GATE** Tests pass; packet capture saved	**GATE** Three consecutive clean demo runs

IPv6 and SRP stay outside the gate

Bibliography

[1] S. Cheshire and M. Krochmal, “Multicast DNS,” RFC 6762, IETF, Feb. 2013.
https://www.rfc-editor.org/info/rfc6762

[2] S. Cheshire and M. Krochmal, “DNS-Based Service Discovery,” RFC 6763, IETF, Feb. 2013.
https://www.rfc-editor.org/info/rfc6763

[3] K. Lynn, S. Cheshire, M. Blanchet, and D. Migault, “Requirements for Scalable DNS-Based Service Discovery (DNS-SD) / Multicast DNS (mDNS) Extensions,” RFC 7558, IETF, Jul. 2015.
https://www.rfc-editor.org/info/rfc7558

[4] S. Cheshire, “Discovery Proxy for Multicast DNS-Based Service Discovery,” RFC 8766, IETF, Jun. 2020.
https://www.rfc-editor.org/info/rfc8766

[5] T. Lemon and S. Cheshire, “Service Registration Protocol for DNS-Based Service Discovery,” RFC 9665, IETF, Jun. 2025.
https://www.rfc-editor.org/info/rfc9665

[6] python-zeroconf project, official repository and documentation (tooling reference).
https://github.com/python-zeroconf/python-zeroconf

[7] dnspython project, official documentation (tooling reference).
https://dnspython.readthedocs.io

[8] Wireshark Foundation, Wireshark official documentation (tooling reference).
https://www.wireshark.org/docs/

Standards metadata checked against the RFC Editor index. Tooling references support the implementation stack shown on slide 7.

Thank You for Your Attention!

NIRIKSHA

“Discover. Examine. Verify.”

From service discovery to verified observation.



Advertised state → Examination → Verified observation

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